

# **STA 131A: Introduction to Probability Theory**

## **Lecture 21: Moment Generating Functions**

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Spring 2026, UC Davis

# Announcements

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## Midterm 2

- Solutions and scores are posted online
- Please review the solution carefully and identify topics to revisit
- You may review your graded exam in discussion section tomorrow
- Regrade requests, if any, should follow the instructions posted on Canvas

## Homework 6 is posted

- This includes re-working Midterm 2 problems and reflecting on what you might have missed and what you have reviewed
- Start early and post questions on Piazza as needed

## Office hours today

- 2:30–3:30 PM at MSB 4220

# Agenda

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## Last time:

- Covariance and correlation
- Variance of sums
- Conditional variance

## Today:

- **Variance decomposition:** finishing the story of conditional variance
  - Law of total variance
  - Estimation/regression interpretation
- **Moment generating functions:** packaging moments into one function
  - Definition and examples
  - Computing moments from MGFs
  - Identifying distributions from MGFs

## Recap: Conditional variance

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### Definition

The **conditional variance** of  $X$  given  $Y = y$  is

$$\text{Var}(X \mid Y = y) = \mathbb{E} \left[ (X - \mathbb{E}[X \mid Y = y])^2 \mid Y = y \right].$$

Equivalently,

$$\text{Var}(X \mid Y = y) = \mathbb{E}[X^2 \mid Y = y] - (\mathbb{E}[X \mid Y = y])^2.$$

- $\text{Var}(X \mid Y = y)$  quantifies the remaining uncertainty in  $X$  after observing  $Y = y$
- $\text{Var}(X \mid Y)$  is a random variable that takes value  $\text{Var}(X \mid Y = y)$  when  $Y = y$

**Question:** How does the “total uncertainty”  $\text{Var}(X)$  relate to the collection of conditional variances  $\text{Var}(X \mid Y = y)$ ?

# Law of total variance

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## Theorem (Law of total variance)

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]).$$

- $\mathbb{E}[\text{Var}(X \mid Y)]$ : average uncertainty remaining after observing  $Y$ .
- $\text{Var}(\mathbb{E}[X \mid Y])$ : variability explained by how the conditional mean changes with  $Y$ .

Interpretation:

total variability = average within-condition variability + between-condition variability.

This is the variance analogue of the law of total expectation:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]].$$

## Proof: Why the law of total variance holds

### Theorem (Law of total variance)

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]).$$

**Proof.** Let  $m(Y) = \mathbb{E}[X \mid Y]$ . Then

$$\text{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - m(Y)^2.$$

Taking expectations and using the law of total expectation,

$$\mathbb{E}[\text{Var}(X \mid Y)] = \mathbb{E}[\mathbb{E}[X^2 \mid Y]] - \mathbb{E}[m(Y)^2] = \mathbb{E}[X^2] - \mathbb{E}[m(Y)^2]. \quad (1)$$

Also, since  $\mathbb{E}[m(Y)] = \mathbb{E}[X]$  by total expectation,

$$\text{Var}(\mathbb{E}[X \mid Y]) = \text{Var}(m(Y)) = \mathbb{E}[m(Y)^2] - (\mathbb{E}[m(Y)])^2 = \mathbb{E}[m(Y)^2] - (\mathbb{E}[X])^2. \quad (2)$$

Adding the two equations (1) and (2) gives

$$\mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \text{Var}(X).$$

## Example: Total variance in the two-group model

### Example

Recall the two-group model:

$$Y \sim \text{Bernoulli}(1/2), \quad X \mid Y = 0 \sim N(0, 1), \quad X \mid Y = 1 \sim N(2, 4).$$

Thus,

$$\mathbb{E}[X \mid Y] = 2Y, \quad \text{Var}(X \mid Y) = 1 + 3Y.$$

Since  $Y \sim \text{Bernoulli}(1/2)$ ,

$$\mathbb{E}[\text{Var}(X \mid Y)] = \mathbb{E}[1 + 3Y] = 1 + 3 \cdot \frac{1}{2} = \frac{5}{2}.$$

Also,

$$\text{Var}(\mathbb{E}[X \mid Y]) = \text{Var}(2Y) = 4\text{Var}(Y) = 4 \cdot \frac{1}{4} = 1.$$

Therefore, by the law of total variance,

$$\text{Var}(X) = \frac{5}{2} + 1 = \frac{7}{2}.$$

## Estimation viewpoint: Resolved vs. remaining variability

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**Inverse-problem setting:** Suppose  $X$  is an unknown quantity of interest, and  $Y$  is an observed measurement/data. Before observing  $Y$ , uncertainty about  $X$  is measured by  $\text{Var}(X)$ .

**Estimation of  $X$  given  $Y$ :** After observing  $Y$ , a natural prediction of  $X$  is the conditional mean

$$\hat{X} = \mathbb{E}[X \mid Y].$$

Define the residual  $R = X - \hat{X}$ . Then

$$\mathbb{E}[R \mid Y] = \mathbb{E}[X - \mathbb{E}[X \mid Y] \mid Y] = 0.$$

Thus, after using the information in  $Y$ , the residual has conditional mean zero.

Moreover, since  $\mathbb{E}[R] = 0$ ,  $\text{Var}(R) = \mathbb{E}[R^2]$ . Therefore,

$$\text{Var}(\hat{X}) = \text{Var}(\mathbb{E}[X \mid Y]) \quad \text{and} \quad \text{Var}(R) = \mathbb{E}[R^2] = \mathbb{E}[\text{Var}(X \mid Y)].$$

Therefore, the law of total variance can be read as

$$\text{Var}(X) = \underbrace{\text{Var}(\hat{X})}_{\text{uncertainty resolved/explained by } Y} + \underbrace{\text{Var}(R)}_{\text{average posterior uncertainty left after } Y}.$$

**Message:** Observing  $Y$  changes the conditional mean  $\mathbb{E}[X \mid Y]$ . The variability of these conditional means is the explained part; the residual term is the average uncertainty left after observing  $Y$ .



# Regression viewpoint: Explained vs. residual variability

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**Regression setting:** Suppose now that

$$Y = \text{response}, \quad X = \text{predictor}.$$

The conditional mean

$$m(X) = \mathbb{E}[Y \mid X]$$

is the regression function: it gives the mean response after seeing the predictor value.

The law of total variance becomes

$$\text{Var}(Y) = \underbrace{\text{Var}(m(X))}_{\text{variation explained by the predictor}} + \underbrace{\mathbb{E}[\text{Var}(Y \mid X)]}_{\text{average within-predictor noise}}.$$

**Interpretation:**

- If  $X$  is informative, then  $m(X)$  changes substantially with  $X$ , so  $\text{Var}(m(X))$  is large.
- If responses are tightly concentrated around  $m(X)$ , then  $\mathbb{E}[\text{Var}(Y \mid X)]$  is small.

This is the probability-theory version of the familiar idea:

$$\text{total variation} = \text{explained variation} + \text{residual variation}.$$

## Pop-up quiz: Total variance

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Suppose that

$$\text{Var}(X \mid Y) = 2 \quad \text{for every value of } Y, \quad \mathbb{E}[X \mid Y] = 3Y, \quad \text{Var}(Y) = 4.$$

**Question:** What is  $\text{Var}(X)$ ?

- A) 2
- B) 6
- C) 14
- D) 38

**Answer: D.**

By the law of total variance,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]) = 2 + \text{Var}(3Y) = 2 + 9 \cdot 4 = 38.$$

**Follow-up:** In

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]),$$

which term becomes small if observing  $Y$  makes  $X$  almost deterministic?

# Motivation for moment generating functions

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We have computed moments directly:

$$\mathbb{E}[X], \quad \mathbb{E}[X^2], \quad \text{Var}(X).$$

But repeating direct integration or summation for each can be tedious.

A **moment generating function** packages all moments into a single function:

$$M_X(t) = \mathbb{E}[e^{tX}].$$

Why is this useful?

- Differentiating  $M_X$  gives moments
- MGFs can identify distributions
- They can make sums of independent random variables easier; we will use this next lecture

# Moment generating function

## Definition (Moment generating function)

The **moment generating function** (or **MGF**) of a random variable  $X$  is

$$M_X(t) = \mathbb{E}[e^{tX}],$$

for values of  $t \in \mathbb{R}$  where this expectation is finite.

- Discrete case:

$$M_X(t) = \sum_x e^{tx} p_X(x).$$

- Continuous case:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx.$$

The MGF is a function of the parameter  $t$ , not a single number.

- $M_X(0) = 1$  always.
- To be useful as an MGF,  $M_X(t)$  should be finite on an open interval around 0.
- Some random variables do not have a finite MGF near 0.

## Examples: Bernoulli and Poisson MGFs

### Example (Bernoulli MGF)

Let  $X \sim \text{Bernoulli}(p)$ . Then

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

Thus,

$$M_X(t) = \mathbb{E}[e^{tX}] = (1 - p)e^{t \cdot 0} + pe^{t \cdot 1} = 1 - p + pe^t.$$

### Example (Poisson MGF)

Let  $X \sim \text{Poisson}(\lambda)$ . Then

$$\begin{aligned} M_X(t) &= \mathbb{E}[e^{tX}] = \sum_{k=0}^{\infty} e^{tk} e^{-\lambda} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = \exp[\lambda(e^t - 1)]. \end{aligned}$$

## Example: Exponential MGF

### Example

Let  $X \sim \text{Exponential}(\lambda)$ , so

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

For  $t < \lambda$ ,

$$\begin{aligned} M_X(t) &= \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx \\ &= \lambda \int_0^{\infty} e^{-(\lambda-t)x} dx \\ &= \frac{\lambda}{\lambda - t}. \end{aligned}$$

For  $t \geq \lambda$ , the integral diverges. Thus the MGF for  $\text{Exponential}(\lambda)$  exists only for  $t < \lambda$ .

## Example: Standard normal MGF

### Example (Standard normal MGF)

Let  $Z \sim N(0, 1)$ . Then

$$\begin{aligned}M_Z(t) &= \int_{-\infty}^{\infty} e^{tz} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp \left[ -\frac{(z-t)^2}{2} + \frac{t^2}{2} \right] dz \\&= \exp \left( \frac{t^2}{2} \right) \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp \left[ -\frac{(z-t)^2}{2} \right] dz}_{=1, \text{ normal density with mean } t} \\&= e^{t^2/2}.\end{aligned}$$

# MGFs generate moments

## From MGFs to moments

When the MGF exists in an interval around 0, differentiating under the expectation gives

$$M_X^{(k)}(0) = \left. \frac{d^k}{dt^k} M_X(t) \right|_{t=0} = \mathbb{E}[X^k].$$

In particular,

$$M'_X(t) = \mathbb{E}[Xe^{tX}], \quad M''_X(t) = \mathbb{E}[X^2 e^{tX}].$$

Setting  $t = 0$ ,

$$M'_X(0) = \mathbb{E}[X], \quad M''_X(0) = \mathbb{E}[X^2].$$

Therefore,

$$\text{Var}(X) = M''_X(0) - (M'_X(0))^2.$$

**Takeaway:** With MGF, all moments can be computed via differentiation.



## Example: Moments from Bernoulli MGFs

### Example (Bernoulli)

For  $X \sim \text{Bernoulli}(p)$ ,

$$M_X(t) = 1 - p + pe^t.$$

Thus,

$$M'_X(t) = pe^t, \quad M''_X(t) = pe^t.$$

Therefore,

$$\mathbb{E}[X] = M'_X(0) = p$$

and

$$\begin{aligned} \text{Var}(X) &= M''_X(0) - (M'_X(0))^2 \\ &= p - p^2 \\ &= p(1 - p). \end{aligned}$$

## Example: Moments from Poisson MGFs

### Example (Poisson)

For  $X \sim \text{Poisson}(\lambda)$ ,

$$M_X(t) = \exp[\lambda(e^t - 1)].$$

Differentiating,

$$M'_X(t) = \lambda e^t M_X(t),$$

and

$$M''_X(t) = (\lambda e^t + \lambda^2 e^{2t}) M_X(t).$$

Thus,

$$M'_X(0) = \lambda, \quad M''_X(0) = \lambda + \lambda^2.$$

Hence

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

## Example: Moments from the exponential MGF

### Example

If  $X \sim \text{Exponential}(\lambda)$ , then

$$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

Differentiate:

$$M'_X(t) = \frac{\lambda}{(\lambda - t)^2}, \quad M''_X(t) = \frac{2\lambda}{(\lambda - t)^3}.$$

Thus,

$$\mathbb{E}[X] = M'_X(0) = \frac{1}{\lambda}, \quad \mathbb{E}[X^2] = M''_X(0) = \frac{2}{\lambda^2}.$$

Therefore,

$$\text{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

## Pop-up quiz: Moments from an MGF

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Suppose a random variable  $X$  has MGF

$$M_X(t) = \exp(2t + 3t^2).$$

**Question:** What are  $\mathbb{E}[X]$  and  $\text{Var}(X)$ ?

- A)  $\mathbb{E}[X] = 2, \quad \text{Var}(X) = 3$
- B)  $\mathbb{E}[X] = 2, \quad \text{Var}(X) = 6$
- C)  $\mathbb{E}[X] = 5, \quad \text{Var}(X) = 3$
- D)  $\mathbb{E}[X] = 5, \quad \text{Var}(X) = 6$

**Answer: B.** We have

$$M'_X(t) = (2 + 6t)e^{2t+3t^2}, \quad \text{and thus,} \quad M'_X(0) = 2.$$

Similarly,

$$M''_X(t) = [6 + (2 + 6t)^2]e^{2t+3t^2}, \quad \text{and thus,} \quad M''_X(0) = 10.$$

Therefore,

$$\text{Var}(X) = M''_X(0) - (M'_X(0))^2 = 10 - 4 = 6.$$

# MGFs can identify distributions

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## Inversion property

If  $M_X(t)$  is finite for all  $t$  in some open interval  $(-a, a)$  around 0, then the MGF  $M_X$  uniquely determines the distribution (CDF) of  $X$ .

If two random variables have the same MGF on an open interval around  $t = 0$ , then they have the same distribution.

## Important remarks:

- The uniqueness statement requires the MGF to exist near 0.
- There exist explicit inverse-transform formulas that allow us to recover the PMF or PDF
- However, we usually identify distributions by “pattern matching” to known MGFs.

## Examples: Identifying distributions from MGFs

### Example

If

$$M_X(t) = \exp\{\lambda(e^t - 1)\},$$

then  $X \sim \text{Poisson}(\lambda)$ , because this is the MGF of a Poisson random variable.

### Example

If

$$M_X(t) = (1 - p + pe^t)^n,$$

then  $X \sim \text{Binomial}(n, p)$ , because this is the MGF of a binomial random variable.

To verify this, we can compute the MGF of  $Y \sim \text{Binomial}(n, p)$ :

$$M_Y(t) = \mathbb{E}[e^{tY}] = \sum_{y=0}^n e^{ty} p_Y(y) = \sum_{y=0}^n e^{ty} \binom{n}{y} p^y (1-p)^{n-y}.$$

## MGFs of independent sums

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If  $X$  and  $Y$  are independent, then

$$\begin{aligned}M_{X+Y}(t) &= \mathbb{E}[e^{t(X+Y)}] \\&= \mathbb{E}[e^{tX} e^{tY}] \\&= \mathbb{E}[e^{tX}] \mathbb{E}[e^{tY}] && \text{(by independence)} \\&= M_X(t) M_Y(t).\end{aligned}$$

This is the transform version of convolution:

convolution of distributions  $\longleftrightarrow$  multiplication of MGFs.

We will use this next time to analyze sums of random variables.

## Wrap-up

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### Law of total variance

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]).$$

- Average remaining uncertainty plus variability explained by  $Y$

### Moment generating functions

$$M_X(t) = \mathbb{E}[e^{tX}].$$

- If  $M_X$  exists near 0, then  $M_X^{(k)}(0) = \mathbb{E}[X^k]$
- $\text{Var}(X) = M_X''(0) - (M_X'(0))^2$
- MGFs can identify distributions when they exist on an interval around 0

**Next lecture:** MGFs for sums of independent random variables and random sums

*Suggested reading:* [BT08, Ch. 4.3 & 4.4]



# References

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Dimitri Bertsekas and John N Tsitsiklis.

*Introduction to probability*, volume 1.

Athena Scientific, 2nd edition, 2008.